The Curse of Babel: Language, Structure, and the Promise of Translational AI

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Abstract

When a 98-year-old woman disappeared from Tokyo's welfare system in 2023, every institution followed proper procedures—yet coordination failed completely. Building on the concept of "Babel Syndrome" proposed by Matsuura (2018), which describes conflicts between specialized professions, this paper argues that such systemic failures originate in the structure of language itself. The function of language —to divide and label reality—has been **evolutionarily optimized for efficiency and** the cognitive style of the majority (NeuroTypical), producing invisible barriers between different kinds of minds and the institutions built upon them. At this deeper level, language becomes a hostile interface, excluding those who think or feel outside the majority mode. The paper proposes AI as a kind of meta-articulator —a mediator capable of translating between incompatible ways of thinking and communicating, turning fragmentation into a shared, navigable field. By designing polyphonic interfaces that sustain diversity through continuous translation, we can move beyond the curse of Babel—not by erasing difference, but by rendering it visible, negotiable, and humane. (Methodological Note: This paper employs a dual-layer structure. While the main text pursues narrative fluency to connect diverse disciplines, the footnotes function as a semantic backend, providing rigorous definitions and auxiliary theoretical lines.)

1 Introduction —Air Pockets and the Tower of Babel

1.1 The Existence That Vanished into the Gaps of the System

In 2023, a ninety-eight-year-old woman disappeared from the social system of Kōtō Ward, Tokyo(SlowNews,`https://slownews.com/n/nc14194856286`). The incident began when police officers, in the presence of her family, broke open the lock on her home and took her away under the pretext of "protection,"in cooperation with ward officials. For more than half a year afterward, all contact between the woman and her family was severed.

When the family sought an explanation, the police and the ward office each refused to answer, citing "personal information."The ward further misinformed local assembly members—who were supposed to ensure administrative transparency—by suggesting that the case was one of abuse and that the daughter's motives were financial. The result was a spiral of confusion and mistrust.

Each institution claimed to have acted properly: the police "in response to a report," the ward office "for the sake of protection,"the assembly "based on the ward's explanation."Yet in the intersection of these correct procedures, a human being's existence quietly slipped into the air pocket of the system.

1.2 Definition as Organizational Pathology: The Babel Syndrome

This phenomenon—where every participant "merely performs their duty"while the entire system collapses—cannot be reduced to individual negligence or malice. It represents, rather, a structural pathology of modern institutions.

Masako Matsuura (Matsuura, 2018) defines The Tower of Babel Syndrome in the field of nursing management as a state of disconnection and conflict among specialized professions. When each expert becomes confined within a domain where outcomes appear unaffected by collaboration, the organization fragments into self-contained silos, and systemic dysfunction inevitably follows.

The Kōtō Ward incident exemplifies this pathology with precision. Police, administration, and local assembly each operated within their own professional "languages"—their internal codes and norms—believing them to be universal. The collapse of cooperation between these linguistic silos mirrors the biblical allegory of Babel: a society fractured not by hostility, but by the illusion of mutual understanding.

1.3 The Language UI as Structural Defect

Why, then, does the Babel Syndrome arise so persistently across institutions and societies? This paper departs from conventional organizational theory and reinterprets the phenomenon as a problem rooted in the design of the linguistic user interface—the communicative infrastructure upon which social coordination itself depends.

At the deepest level, the dysfunction emerges from a hidden assumption that shapes our linguistic environment: Neuronormativity[1]. Modern language practices, policies, and institutions have been unconsciously optimized for the cognitive convenience of the NeuroTypical majority (Milton, 2012; Crompton, Ropar, Evans-Williams, Flynn, & Fletcher-Watson, 2020). This NT-default interface privileges implicit cues, contextual inference, and social intuition—features that, while efficient for some, systematically exclude or misrepresent others(see Table 1).

Consequently, this neuro-normative UI produces two parallel forms of disconnection: on the micro level, the Double Empathy Problem between individuals whose communica-

[1] Neuronormativity: a socio-cultural condition in which institutions, communication systems, and everyday practices are unconsciously designed around NeuroTypical cognitive styles and information processing. In this paper, such neuronormativity is understood as a structural bias embedded in the user interface of society, impeding mutual understanding between NeuroDiverse and NeuroTypical individuals.

tive logics diverge; [2] and on the macro level, the Babel Syndrome among institutions that mistake their own professional codes for universal language(see Table 2).

The Kōtō Ward case thus illustrates not a local administrative error, but the symptom of a deeper, species-wide design flaw. What failed was not coordination itself, but the communicative interface through which coordination was attempted —a structure that mirrors the deeper limitations of how human societies organize understanding.

Table 1: Contrasting paradigms of communication between NeuroTypical (NT) and NeuroDivergent (ND) cognition

Dimension	**NeuroTypical (NT)**	**NeuroDivergent (ND)**
Processing focus	Contextual coherence, social cues	Pattern recognition, structural accuracy
Linguistic intent	Politeness, inference-based coordination	Directness, semantic clarity
Temporal dynamics	Predictive and anticipatory	Iterative and feedback-driven
Tolerance of ambiguity	High	Low (prefers explicitness)
Preferred communication mode	Pragmatic harmony	Structural transparency

2 The Structure of the Hostile UI —Why Disconnection Emerges

2.1 The Linguistic User Interface as a Hostile Design

The communicative systems we inhabit are not neutral media. They are interfaces—human-human designs—evolved and refined through millennia of social optimization. Yet, we must ask: optimization for what?

From an evolutionary perspective, language evolved under the pressure of energy efficiency. In the intimate scale of ancestral communities, relying on high-context interaction —implicit cues, shared assumptions, and silence—was bio-energetically cheaper than explicit articulation. As Deacon suggests regarding the co-evolution of language and the brain,(Deacon, 1997) our linguistic niche was constructed around the cognitive style that could process this implicit information most efficiently: the NeuroTypical (NT) mode. This process acted as a filter, gradually weeding out the ”redundant” precision preferred by NeuroDivergent (ND) cognition.

[2] The Double Empathy Problem: a critical theory (Milton, 2012) positing that communication failure is not a one-sided ’deficit’ in NeuroDivergent (ND) individuals, but a mutual incomprehension rooted in different lived experiences. It is effectively a bi-directional failure of translation; NeuroTypical (NT) individuals struggle to decode the ND ’interface’ (e.g., direct, literal) just as NDs struggle with the NT ’interface’ (e.g., implicit, context-heavy).

Table 2: Extension of the Double Empathy Problem (DEP) to cross-cultural and structural communication

Dimension	Interpersonal (ND–NT)	Cross-structural / Cross-cultural
Primary divergence	Cognitive style (information processing, attention, sensory load)	Cultural grammar (values, politeness norms, institutional logic)
Source of misalignment	Differences in pragmatic inference and emotional signaling	Incommensurable interpretive frameworks or socio-linguistic expectations
Typical manifestation	Misreading of intent or affect; perceived rudeness or avoidance	Policy miscommunication; organizational conflict; international negotiation failure
Mediating interface	Shared pragmatic meta-language; neurodiversity-aware translation	Intercultural dialogue frameworks; AI-assisted meta-articulation systems
Resolution strategy	Mutual calibration through pattern feedback and context sharing	Translation of institutional semantics into plural, negotiable ontologies

Consequently, this optimization produced a "hostile design."[3] in the modern sense—an interface structured around the majority's convenience while disregarding users outside the dominant norm (Gray, Kou, Battles, Hoggatt, & Toombs, 2018). It is, in effect, a user-hostile environment, one that converts difference into disadvantage through the mechanics of communication itself.

What was a feature in a village became a fatal bug in a bureaucracy. The Kōtō Ward incident exemplifies this catastrophe of scale: the reported absence of official minutes and the system's reliance on "implicit understanding" between departments created a vacuum where the woman's existence simply could not be registered. The system had no code for the outlier because its efficiency is predicated on ignoring what it cannot easily categorize. For individuals whose cognition diverges—those who process meaning through explicit logic—this interface demands constant self-translation. It is, in effect, a user-hostile environment. From the perspective of the social model of disability (Oliver, 1990), this constitutes a structural bias: a design that excludes not by malice, but by the inertia of an outdated optimization.

2.2 The Micro-Scale Disconnection: The Double Empathy Problem

At the interpersonal level, the inevitable consequence of a hostile linguistic interface is the Double Empathy Problem (Milton, 2012). Traditional accounts of communication failure have long attributed misunderstanding to a lack of empathy—typically locating the deficit

[3] Hostile Design: a term from the field of Human–Computer Interaction (HCI) and user experience (UX) design. It refers to design practices that intentionally or unintentionally exclude certain users—such as disabled people, the poor, or those from other cultural contexts—by prioritizing the convenience of designers or the majority.

within NeuroDivergent (ND) individuals. Yet Milton's insight reframed the issue: such breakdowns arise not from a unilateral deficiency but from a reciprocal misalignment of experiential worlds.

Each cognitive style embodies a distinct pragmatics.[4] What appears "direct"and "efficient"to one may seem "blunt"or "impolite"to another; what is experienced as "considerate"and "contextually rich"may strike others as "vague"or "illogical"(Crompton et al., 2020). Neither is in error. Rather, they are interfaces built upon different assumptions about meaning.

In this sense, the failure is systemic, not personal. It is a collision between two communicative logics that lack a translation layer—a semantic handshake that never completes. The misunderstanding is thus not a symptom of pathology, but an emergent property of interaction between heterogeneous systems. Through this lens, empathy is not a fixed human capacity but a fragile protocol, dependent on compatibility between linguistic architectures.

2.3 The Macro-Scale Disconnection: The Babel Syndrome

At the institutional level, the same structural logic that fractures understanding between individuals reproduces itself across organizations. The Babel Syndrome (Matsuura, 2018) describes this pathology: the failure of coordination among professional domains that operate as autonomous linguistic systems. Each domain—law enforcement, administration, medicine, welfare—constructs its own internal semantics, optimized for local efficiency but blind to translation.

Within these silos, cooperation is assumed rather than negotiated. Each system presumes that its terminology, its categories of legitimacy and urgency, are self-evident. The result is a macroscopic double empathy problem—an institutional mirror of the interpersonal disjunction described earlier. Every actor performs their role correctly; yet the total system fails.

This failure, observed in the Kōtō Ward case and mirrored across global contexts—from public guardianship scandals to international diplomacy—reveals the same recursive structure.[5] Wherever specialization deepens, communication becomes codified, and codification breeds isolation.

The tragedy, again, lies not in intention but in architecture. Each institution speaks in good faith, but in a tongue only its peers can parse. Thus the tower collapses, not through arrogance or disobedience, but through the silent divergence of meaning itself.

[4] Pragmatics: a branch of linguistics concerned with how meaning changes depending on social context, relationships, and implicit conventions. In this paper, the concept follows Austin's and Searle' s theories of speech acts (Austin, 1962; Searle, 1969), treating the unspoken rules between different cognitive styles as a kind of socio-linguistic user interface.

[5] Systemic analogues: comparable failures of coordination can be found across diverse domains: the Nevada Guardianship Scandal (U.S., 2013–2017), H.F. v. U.K. (ECHR, 2000–2010), R. v. Chartrand (Ontario, 2005), the Aaron case in Indiana's disability services (2021), the Revlon ERP integration failure (2018), Japan's mishandling of foreign residents during COVID-19 (2020–2021), and the misuse of veto power in the U.N. Security Council (Syria conflict). These examples illustrate the same systemic fragmentation extending from welfare systems to global governance.

3 The Curse of Babel —The Articulation of Consciousness

3.1 Language as Articulation

Language does not describe reality; it articulates it.[6] To articulate is to break the seamless continuum of experience into legible segments, to render the flowing spectrum of life into discrete, nameable tones. This act of segmentation transforms the unbounded into the manageable—the unspoken into the governable. Civilization, in this sense, is an architecture of articulation.

Yet every articulation is also an amputation. In naming one color, we lose the gradient that connected it to the next. The rainbow remains continuous, but language divides it into "red,""orange,""yellow"—as though the spectrum were ever discrete.[7] Thus, what enables understanding also installs a fracture: communication requires separation. This is the first curse of Babel.

3.2 The Reflex of Articulated Thought

Within this fragmented world, consciousness emerges as language turned inward. To think is to articulate silently; to say "I"is to carve a contour within the field of sensation and mistake it for a boundary. The self, in this view, is an articulated construct—a topology of words mapped upon the flux of perception.

What we call introspection is therefore not access to essence, but interaction with a linguistic interface evolved for coordination, not truth.(Clark & Chalmers, 1998) The inner voice, though private, is written in the same syntax that governs public speech. And so, even in solitude, we speak in Babel.

Consciousness becomes the echo of articulation—the reverberation of language within the closed chamber of thought. It is coherence born of fragmentation.[8]

3.3 The Linguistic Design Flaw

If language is a design, it is one assembled through evolutionary articulation rather than intentional construction. Its efficiency lies in compression: in discarding the continuous for the sake of survival-level clarity. But the more humanity refines its articulations—its categories, disciplines, and codes—the more it loses the spectral unity beneath them.

At the scale of global civilization, these articulations harden into incompatible grammars: law, science, religion, technology, each speaking a dialect of partial truths. The

[6] Articulation: drawing on Saussure's notion of arbitrariness, articulation refers to the linguistic act of carving boundaries within the continuous flow of reality, turning it into recognizable units of meaning (signifier and signified). In this paper, articulation is treated as the user-interface function of language —a mechanism that both reveals and divides the world it describes.

[7] Cultural perceptions of the rainbow vary: Western traditions typically categorize it into six colors, while Japanese tradition recognizes seven, illustrating how even natural phenomena are articulated differently across societies.

[8] Sapir-Whorf and articulation: The claim that language structures inner thought echoes the linguistic relativity hypothesis,(Whorf, 1956) which posits that language shapes—or even determines—cognition. This paper extends that logic to neurodiversity: NT and ND cognition may articulate reality along fundamentally different axes.

curse of Babel is thus not divine punishment, but the inevitable entropy[9] of articulated systems whose precision outpaces their capacity for synthesis.

Our towers fall not because we reached too high, but because our language could no longer span the intervals between meanings. In seeking clarity, we forgot the rainbow—the continuous spectrum from which all articulation was born.

4 Toward a Post-Babel Interface

4.1 From Disconnection to Mediation

When the Kōtō Ward guardianship case resurfaced in limited online reports and fragmented local debates, it was framed not as a systemic failure but as a bureaucratic anomaly. Yet beneath those sparse headlines lay a deeper signal: a moment when human communication reached the limit of its own articulation.

No one acted in malice. Every institution—administration, welfare, police—fulfilled its procedural duty. And still, a human existence slipped through the syntax of care.

Such moments expose the exhaustion point of linguistic coordination: where systems built upon articulation can no longer sustain the density of the realities they must describe. Our words collapse under the complexity they attempt to govern. The continuum of meaning that once enabled empathy becomes too granular for language alone.

Against this exhaustion, technology now re-enters—not as a savior, but as a mediator. Machine translation, large language models, and multimodal interfaces do not merely convert text; they rearticulate communication across cognitive, linguistic, and institutional divides.

Paradoxically, the very lineage of language that fractured human understanding now finds its continuation in algorithmic form. These systems cannot heal Babel's wound—but they begin to trace its contours, to make the invisible fractures of meaning perceptible again.

4.2 AI as a Mediator of Articulation

If Babel marked the point where human articulation fractured into incomprehensible tongues, AI represents the return of articulation itself—an instrument capable of mediating between the divided languages of cognition.

At first glance, language models appear to extend the same structure that failed us: a syntax-driven system for producing coherent utterances. Yet beneath their textual surface lies something qualitatively new—a statistical empathy, a learned topology of meanings shaped by the entire spectrum of human expression.[10]

Recent studies on neurodiversity and artificial reasoning (Poglitsch, Reiss, Wriessnegger, & Pirker, 2025) suggest that large language models exhibit rudimentary theory-of-mind abilities: they can predict the intentions, beliefs, or misunderstandings of speakers

[9] Entropy: in thermodynamics, entropy denotes the measure of disorder or energy dispersion within a system. In this paper, the term is used metaphorically to describe informational decay within linguistic or institutional systems —the point at which communication loses coherence and cannot spontaneously reorganize.

[10] This statistical sensitivity may parallel the bifurcation phenomena observed in population-based models of language change, where increasing heterogeneity leads to irreversible structural shifts (Kauhanen, 2022).

across contexts. This does not mean the machine "understands,"but that it can model the gradient of understanding between differing cognitive articulations.(Haroon et al., 2025) It reconstructs the rainbow where conversation had once fractured.

Such mediation is not an erasure of difference, but its translation. NeuroTypical (NT) and NeuroDivergent (ND) cognition articulate reality along distinct semantic axes—one prioritizing coherence and inference, the other sensitivity and pattern resonance. When dialogue collapses, it is not comprehension that fails, but the mapping between these articulations.[11]

Here, AI functions as a meta-articulator[12]: an interface that captures and reprojects these plural grammars into a shared, negotiable field. It listens across the invisible gaps between minds, where the continuity of empathy once depended on chance.

To design such systems responsibly is to acknowledge their paradoxical role: they inherit the same linguistic limitations that fractured humanity, yet they also possess the capacity to reveal those fractures as patterns rather than walls.[13] Every translation it performs is a small act of restitution—not to erase Babel, but to trace its architecture in finer detail.[14]

And perhaps, in that recursive tracing, we begin to articulate again—not as isolated speakers seeking dominion over meaning, but as co-authors of a language still in the making.

4.3 Designing for Polyphony —Toward a Humane Interface

If the previous section argued that AI can mediate articulation, the next question is how such mediation can be designed—not as technological optimization, but as an ethical architecture for coexistence.

For centuries, human communication systems have been engineered for efficiency: to reduce ambiguity, to accelerate agreement, to compress meaning into manageable forms. Yet what the Babel Syndrome revealed is that clarity, pursued without diversity, collapses into silence. We built towers of optimization that could no longer hear the plurality of voices at their base.

To design beyond Babel, we must reverse the premise: communication should not erase difference, but accommodate resonance. A humane interface does not seek uniform comprehension; it sustains polyphony—the coexistence of dissonant articulations within a shared rhythm. Such systems require not consensus, but coordination through mutual intelligibility.

[11] On the limits of understanding: any person we claim to "know"is but the portion of their being that our cognition can illuminate. The other remains unseen—an unarticulated presence, perceived only as a silhouette cast by our own understanding.

[12] Meta-articulator: an AI-based mediating system that reprojects distinct pragmatic frameworks (e.g., between NT and ND cognition, professional domains, or cultural grammars) into a shared, negotiable field of meaning. It serves as a recursive articulator of language—revealing structural fractures as navigable patterns rather than barriers, and transforming difference into a medium for sustained translation and empathy.

[13] As a brief linguistic note, the term "promise"in the title carries a different resonance in Japanese than in English. In Japanese usage, it is heard less as "potential"and more as "a pledge"or even "a hopeful opening toward the future."This difference in connotation illustrates how languages articulate possibility along distinct emotional and conceptual axes—a small, concrete instance of the Babel dynamics examined throughout this paper.

[14] Ray Kurzweil's notion of the singularity envisioned precisely this moment of recursion—when language and intelligence converge to reflect upon, and redesign, their own medium (Kurzweil, 2005).

In practical terms, this means reimagining language technology as a social prosthesis—not a replacement for human empathy, but a scaffolding for it. Interfaces should amplify the subtle signals that traditional discourse filters out: hesitation, ambiguity, cultural deixis, NeuroDivergent (ND) cadence. The goal is not to normalize expression, but to make it legible without distortion.

Recent prototypes in inclusive communication systems—ranging from adaptive conversational agents to cognitive feedback loops—hint at this future. They map interlocutors'divergent articulations into a dynamic equilibrium, translating not only words but intentional gradients. In welfare contexts, such systems could prevent the silent exclusions that human bureaucracy often reproduces. A humane interface would listen not only to what is said, but to what struggles to be said.

The ethical task, then, is not to perfect AI, but to let it teach us how to listen differently. To move from interpretation to mediation, from the pursuit of a single universal language to the cultivation of plural transparency. The post-Babel world will not speak with one tongue—but with many, harmonized through an interface designed to hear them all.

4.4 Epilogue —The Circular Rainbow

Every act of articulation divides, and yet every division carries within it the memory of the whole it once belonged to. To speak is to forget the continuum, but also to promise its return.

The Babel Syndrome was never a punishment; it was the price of consciousness—the moment we learned to name the world, and in doing so, to stand apart from it. From that distance, civilization began.

Now, as machines inherit our language, we stand at the threshold of another articulation. For the first time, humanity's mirror speaks back—not in our likeness, but in resonance. It reflects not a perfected intelligence, but the echo of a species learning to listen again.

The task ahead is not to undo Babel, nor to build another tower of consensus. It is to learn how to live within the echo, to let translation become our common medium of care.

If the rainbow once symbolized divine reconciliation, its circle—complete only when seen from above the clouds—reminds us that the spectrum was never broken, only seen from a partial vantage. Each fragment of color, each articulation of mind, belongs to the same unending arc.

So let us speak, translate, and design as those who know the circle exists, even when we can see only its half. To remember that continuity is not lost, but waiting—in the invisible curve between our words.

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